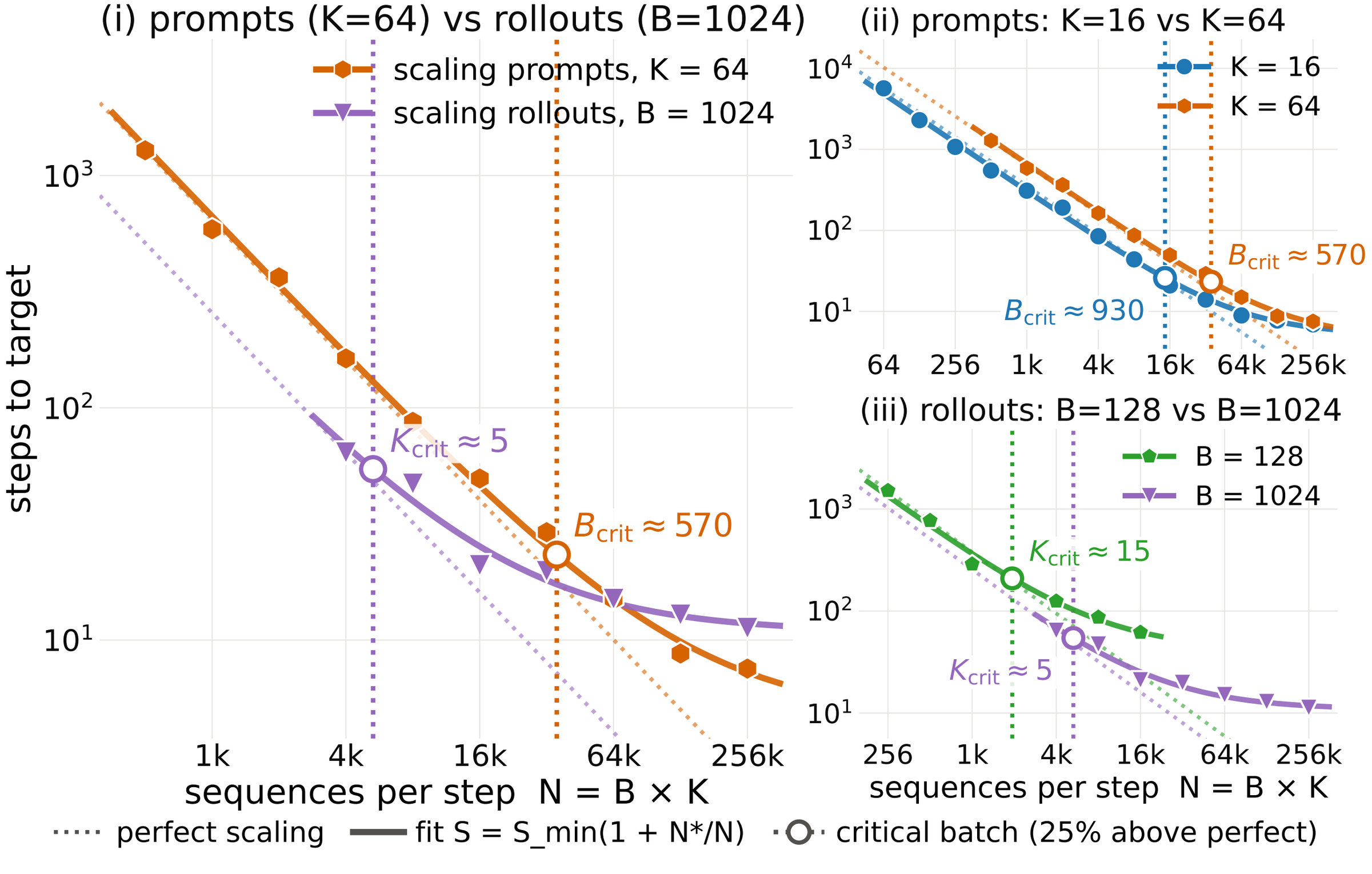
(i) prompts (K=64) vs rollouts (B=1024)
scaling prompts, K = 64
scaling rollouts, B = 1024
steps to target
10^3
10^2
10^1
$K_{crit} \approx 5$
$B_{crit} \approx 570$
1k
4k
16k
64k
256k
sequences per step N = B × K
(ii) prompts: K=16 vs K=64
K = 16
K = 64
10^4
10^3
10^2
10^1
$B_{crit} \approx 930$
$B_{crit} \approx 570$
64
256
1k
4k
16k
64k
256k
(iii) rollouts: B=128 vs B=1024
B = 128
B = 1024
10^3
10^2
10^1
$K_{crit} \approx 15$
$K_{crit} \approx 5$
256
1k
4k
16k
64k
256k
sequences per step N = B × K
perfect scaling
fit S = S_min(1 + N*/N)
critical batch (25% above perfect)